UTSAV CHAUDHARY

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Senior **Full Stack Engineer** with **6+ years** of hands-on **backend** and **frontend** development, plus **AI agents**, **bots**, and **LLM** systems, building production applications in **regulated utility**, **energy**, and **enterprise consulting** environments. Expert in **Python**, **TypeScript**, and **JavaScript** with **Angular**, **React.js**, and **Next.js**, and in **REST APIs** using **FastAPI**, **Flask**, and **Node.js**. Provide **solution design**, development quality, and delivery execution on **AWS** and **Azure** (compute, storage, networking, platform services). Guide teams on **CI/CD**, **Git workflows**, **Terraform**, and **GitHub Actions**, with consistent **SOLID**, **automated testing**, **coding standards**, and **secure coding**.

Technical Skills

Languages & Frontend: Python, TypeScript, JavaScript, C#, Angular, React.js, Next.js, Redux, TailwindCSS, D3.js, Chart.js

Backend & APIs: FastAPI, Flask, Django, ASP.NET, ASP.NET Web API, Node.js (Express), Pydantic, REST APIs, OpenAPI, Swagger, gRPC, WebSockets, GraphQL

AI, LLMs, Agents, & Bots: Azure OpenAI, GPT-4o, GPT-4o-mini, Llama, Hugging Face, RAG, embeddings, vector search, MCP servers, AI agents, Ollama, Whisper, TensorFlow Lite, prompt engineering

Cloud, DevOps, & CI/CD: AWS (EC2, EKS, Lambda, RDS, S3, VPC, IAM, API Gateway, CloudWatch), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Key Vault), Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Bitbucket Pipelines, Git, GitFlow, environment promotion

Networking & Security: VNets, Security Groups, NSGs, DNS, VPN, LDAP, RBAC, SSL, TLS, OAuth 2.0, NERC CIP, ISO 27001, PCI, OWASP, secure coding, vulnerability assessments, enterprise risk controls

Engineering Practices & Tools: OOAD, SOLID, automated testing, code reviews, design reviews, branching strategies, PostgreSQL, MongoDB, Redis, Kafka, Jira

Experience

Sr. Full Stack Engineer / Technical Lead, Hybrid, **MidAmerican Energy**, TX — Feb 2025 to Present

- Provided technical leadership on parallel **AI** and product workstreams for a regulated **electric and natural gas utility**, working with product and architecture leads to turn business requirements into **technical specifications** for **APIs** and UI across **grid telemetry** and **smart meter (AMI)** platform services.
- Led **code reviews**, **design reviews**, and technical validations so assigned workstreams stayed consistent on **architectural principles**, **OOAD**, **SOLID**, **automated testing**, **coding standards**, versioning, and documentation.
- Designed **AI agents**, **bots**, and **MCP servers** that extended **LLM** capabilities with custom tools, file systems, and APIs; owned the manager agent that planned tasks, picked tools, and routed work to specialist agents with human-in-the-loop gates for **outage management** and **metering** workflows.
- Built the shared **Python/TypeScript** tool layer (search, scrape, DB, Gmail) with schemas and retries; wired **Azure OpenAI GPT-4o/GPT-4o-mini** and a **Hugging Face** extraction model as callable tools, with retrieval over **NERC CIP** compliance documents and operational runbooks before cleaned records hit **MongoDB**.
- Built **React.js** and **Next.js TypeScript** applications with **TailwindCSS**, including an internal Admin Portal for KPI tracking, agent-run monitoring, and **outage** and **AMI metering metrics**; translated **Figma** into production UI with **RBAC** and performance-tuned component patterns.
- Developed **Python** microservices and **Node.js REST APIs** over **MongoDB** and **Firebase**; integrated **ERP**, **customer information system (CIS)**, and **market data** feeds with **PCI**-aligned **secure coding** for **customer payments** data.
- Partnered with DevOps on **CI/CD** (**GitHub Actions**, Bitbucket), **Terraform**, **Docker** on **EKS/AKS**, and **environment promotion** across **Dev**, **Test**, **UAT**, and **Prod**; led **root cause analysis** with **Prometheus** and **CloudWatch**.

Full Stack Engineer, On-Site, **Chevron**, GA — Jan 2024 to Jan 2025

- Implemented and reviewed **Python FastAPI/Flask** and **Node.js REST APIs** for **drilling reporting** and **refinery maintenance** workflows, with **automated testing** before **AKS** promotion, helping keep **99.9% availability** behind **Application Gateway** and **Front Door**.
- Delivered **Angular** and **React.js TypeScript** features for **field operations** and maintenance workflow portals with real-time **telemetry** views, keeping UI routing and API contracts intact through **environment promotion** on Azure.
- Performed **code reviews** and **design reviews** on portal and API changes; applied **OOAD**, **SOLID**, and **OWASP** with coding standards and **secure coding**, using **Azure Key Vault** to protect proprietary **seismic and operational data** in **Azure SQL**.
- Helped migrate the portal and **telemetry** stack (**Angular**, **React**, **Python**, **Node.js**) from legacy hosting to cloud native **AKS** on Azure, improving **resource utilization** by **40%** while preserving service boundaries and API contracts.
- Supported canary releases and **environment promotion** through **Azure DevOps** pipelines, **Terraform**, **Git workflows**, and **branching strategies**, contributing to a **75%** reduction in deployment related incidents.
- Contributed to monitoring with **Application Insights**, **Azure Monitor**, and **Grafana**, supporting **MTTD** from **30 minutes** to **5 minutes**; led **root cause analysis** and **vulnerability assessments** with Risk and Compliance.

Full Stack Engineer CO-OP, On-Site, **Midh Technologies**, TX June 2023 to Dec 2023

- Built **Python FastAPI** and **Flask REST APIs** for **Llama** and **Hugging Face** inference behind **AWS API Gateway** / **Azure API Management**, with **IAM** authentication, request throttling, and **Pydantic** validation.
- Developed **React.js** analytics dashboards with **TypeScript**, **D3.js**, and granular **RBAC** for internal teams and client stakeholders, translating analyst requirements into production UI workflows.
- Fine tuned **Llama** models on GPU backed **Kubernetes** (**EKS**/**AKS**) clusters tuned with **node pools**, **resource quotas**, and **auto scaling**; contributed **Python ETL** and **Pandas** jobs that loaded curated tables into **PostgreSQL** and **MongoDB** for dashboard and API consumers.
- Built and deployed the company web platform in **Next.js** and **React** with secure **RBAC** and scalable API integrations on **Azure App Service** and **AWS Amplify**.
- Built production **CI/CD** with **GitHub Actions** and **Azure DevOps**: **pytest** unit tests, Docker image builds, security scanning, **Terraform**, and **environment promotion**.

Software Engineer, Python, On-Site, **Accenture**, IN June 2021 to July 2022

- Built a responsive **React** frontend on a **FastAPI** backend for **enterprise client accounts** in a global consulting engagement, delivering interactive **D3.js** and **Chart.js** metrics dashboards that improved data visualization efficiency by **45%**.
- Optimized **Python FastAPI REST APIs** with **Pydantic** models, sustaining **250ms** average response time behind **Nginx** with **SSL**/**TLS** termination on **hybrid on premise** deployments.
- Built **Python** scraping/**ETL** with **Selenium**, **BeautifulSoup**, and **Pandas**, cutting manual collection time by **60%**; added **WebSocket** streaming so operational dashboards stayed current.
- Implemented self hosted **CI/CD** with **GitHub Actions** and **Jenkins**, **Git branching strategies**, and **Prometheus**/**Grafana** observability; applied **code reviews** and **ISO 27001**-aligned **secure coding**, reducing operational overhead by **60%**.

Software Engineer Intern, Remote, **VRoKCs**, MO March 2020 to May 2021

- Deployed **Python Flask** web applications and **REST APIs** on **Azure App Services** with auto scaling and integrated monitoring.
- Built a **Flask** app (**Python**, **JavaScript**) with **TensorFlow Lite** for real time, in browser sign language model evaluation via image upload and webcam streaming.
- Supported **CI/CD** through **Azure DevOps** and **Git**, using **Azure Blob Storage** for artifacts and **Docker** for inference services.

Web/Game Developer Intern, On-Site, **underDogs Gaming Studios**, IN Jan 2018 to Feb 2019

- Built game service **REST APIs** in **Python** and **Node.js**, integrating the **Quarters API** for wallets and secure transactions across multiple titles.
- Wrote **Python** tooling and **JavaScript** admin screens for payments, inventory, and shipping, keeping client/server flows reliable under failure conditions.
- Supported release and hosting for game backends with **API** rate limiting, health checks, and **Git** source control.

Certifications









Microsoft Certified: DevOps Engineer Expert View Credential
AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python (Specialization) View Credential

Education

State University of New York at Binghamton Aug 2022 to Dec 2023
Master of Science in Computer Science 3.4/4.00 GPA

Universal College of Engineering, Mumbai July 2017 to June 2021
Bachelor of Engineering in Computer Engineering CGPA 8.09/10